

It's for your own good

THIS MONTH WE SAW REPORTS OF FIRMWARE OF MOTHERBOARDS BEING COMPROMISED SINCE 2016. Imagine that, the malware in question was present in the UEFI firmware of some of the most popular motherboard manufacturers. And these manufacturers don't serve just the DIY PC segment but also serve ODMs who make laptops incorporating these motherboards. So we don't know the full extent of the impact of this attack.

The UEFI firmware is what boots the first when you turn on a computer. It runs before everything and makes way for the operating system to start off. So there is no real way of detecting this. Also, there's no real way for antivirus software to fight it since messing with the UEFI firmware could make your computer unbootable. In a way, you have to give it to the folks who executed this attack because they actually came up with a method that's extremely difficult to combat. Security researchers are still trying to figure out how the attackers managed to compromise the UEFI firmware. My bets are that they managed to gain privileged access to the FTP servers where these firmware files are hosted. And obviously, the manufacturers are at fault for not running hash checks on the firmware. That would have caught the modified firmware right before it got flashed onto hundreds of thousand of motherboards. In case you missed it, they've been undetected since 2016.

UEFI started out as a massive improvement over the legacy firmware that preceded it. It not only allowed for a more user friendly interface but the fact that firmware could be much more feature rich has helped the PC ecosystem evolve by leaps and bounds. It's the unfortunate side effect of having too many features. They all end up being additional vectors for malicious actors to compromise. This is true for a lot of new features in the technology space. It's unintentional but inevitable.

What happens when the intentions of introducing a new feature or technology isn't about helping the users or evolving the ecosystem? What if features were introduced without any noble intentions or with the sole purpose of safeguarding your bottom-

line? That's DRM in a nutshell. DRM is often implemented in a very discreet manner. With video games, we tend to discover them when things go awry and games become unplayable. We've all read countless articles of how Denuvo was implemented in video games. They've either made games unplayable because of a nasty performance hit or some unforeseen circumstance that breaks the game altogether and requires a fix. The former only inconveniences people whereas the latter literally means that people are unable to experience the games that they paid for.

DRM isn't always implemented as a software. There are often hardware implementations as well. Microsoft had implemented Secure Boot around the time Windows 8 came out, which had the potential to lock out the user from installing alternative operating systems such as Linux, if they wanted to. Thankfully, there was a flag to disable secure boot within the BIOS to prevent the system from being vendor locked. Apple, on the other hand, has the T2 chip which has been used to prevent not only third-party hardware repairs but also limited the types of operating systems that can be installed on an Apple device. In a way, it's very similar to the Secure Boot system.

A more recent example of a hardware component that can potentially act as a DRM service is Microsoft's Pluton chip. In its current version, it's very much a security solution that's geared to protect consumers from being unsuspecting victims of advanced intrusion attempts. However, the provisions that help Pluton become what it is, also makes way for it to become a gatekeeper of sorts. There are provisions that could prevent machines with Pluton chips from allowing operating systems other than Windows to be installed. This includes Linux which has been evolving to become quite the competition to Windows.

Pluton is very much in its infancy and it's too early to comment how things are going to turn out. It has the potential to be a great security solution but it also has to become the crux of an anti-trust lawsuit. Let's just hope that it doesn't turn out to be for the better. **d**



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